

Multiple Choice (10 marks)

1. Which form of encryption does WPA use?
 - (a) Shared key
 - (b) LEAP
 - (c) TKIP
 - (d) AES
2. How do you prevent SQL injection?
 - (a) Escape queries
 - (b) Interrupt requests
 - (c) Merge tables
 - (d) All of the above
3. A _____ is a method in which a computer security mechanism is bypassed untraceable for accessing the computer or its information.
 - (a) front-door
 - (b) backdoor
 - (c) clickjacking
 - (d) key-logging
4. A _____ may be a hidden part of a program, a separate infected program a Trojan in disguise of an executable or code in the firmware of any system's hardware.
 - (a) crypter
 - (b) virus
 - (c) backdoor
 - (d) key-logger
5. Which of the following does authentication aim to accomplish?
 - (a) Restrict what operations/data the user can access
 - (b) Determine if the user is an attacker
 - (c) Flag the user if he/she misbehaves
 - (d) Determine who the user is

True / False (10 marks)

Answer True or False

6. True or False: The main types of scanning in cybersecurity are network scanning, vulnerability scanning, and port scanning.
7. True or False: The MD 5 algorithm generates a message digest that is 160 bits in length.
8. True or False: A buffer overflow on the stack can overwrite the return address, redirecting execution to attacker-injected code.
9. True or False: DNSlookup is an effective tool for performing SNMP enumeration in network environments.
10. True or False: The one-time pad key can be computed as $k = m \text{ xor } c$, where m is the plaintext message and c is the ciphertext.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a function to find the longest common subsequence for the given two sequences.
12. Write a python function to count unset bits of a given number.

End of Paper